Low-Frequency RoPE Subspace for Near–Far Sparse Attention

Anonymous authors

***Abstract*—Page-to-token sparse attention faces two coupled decisions: how to score a long history cheaply, and how to allocate a limited candidate set before final attention. We study a near–far cascade that uses the same 32 low-frequency RoPE coordinates for page screening and token refinement, reserves page quotas for both regions, and recomputes final attention with original-dimensional queries, keys and values. Qwen3-8B diagnostics show why these decisions interact: regional reservation raises mean far-region attention retention from .6759 to .7207, while reducing it in 43 of 128 correlated observations. The budget analysis further shows that reserving far pages alone need not leave any far-token quota. Historical five-task LongBench scores are 47.588 for shared-subspace scoring and 47.910/47.666 for nominal full-dimensional page/token variants, but do not isolate representation effects. Separate timing studies find no selector-time advantage over nominal full-dimensional page scoring and higher sparse decode time than dense attention. Thus lower scoring dimensionality and regional access are insufficient to establish an efficient sparse decoder. The compact implementation with 4/128 prefix/recent protection still requires matched quality and preparation-inclusive cost evaluation.**

***Index Terms*—sparse attention, shared representation, regional retrieval, long-context inference**

I. Introduction

Long-context decoding repeatedly accesses previously generated keys and values. Sparse attention limits this work by selecting a smaller token set, but selection introduces its own scoring, ranking and memory accesses. A page-to-token cascade adds an early decision: which pages may contribute tokens to refinement? Under sufficient positive-probability candidate support, excluding a page at this stage prevents its unprotected tokens from reaching final attention. The aim is to retain useful candidates within a fixed token budget at a retrieval cost below the attention work saved.

Two observations motivate studying scoring and allocation together (Fig. 1), using the token/page perspective of ClusterKV [7]. First, page summaries change which history is retained: the mass-ranked, mean-score and extrema-score sets differ in the same Qwen3-8B example (a). This comparison mixes representation and pooling choices, so it motivates examining their effect on candidate access rather than identifying a better scorer. Second, a single global budget gives uneven regional coverage (b). Reserving separate near and far capacity increases mean far retention from .6759 to .7207, but decreases it in 43 of 128 correlated observations. Allocation can therefore preserve distant mass at the expense of other selections; it is a resource trade-off, not a per-query guarantee.

These observations lead to a cascade with two explicit controls. A fixed set of complete low-frequency RoPE pairs supplies the scoring coordinates for both stages; separate near–far page quotas determine regional access to token refinement. The coordinate diagnostic in Figure 1(c) motivates the compact view, although its unpaired random baseline does not isolate frequency from pair structure. Original-dimensional Q/K/V are retained for final attention, so approximation enters through token selection. We also consider an optional prefill-informed far gate. The historical layer variation in panel (d) motivates this extension, but uses a different single-query statistic; the gate is disabled by default and must be assessed independently.

We formulate shared-coordinate regional selection and analyze how page and token budgets jointly constrain retrieval. The evaluation examines task quality, regional coverage and execution cost. Qwen3 diagnostics show changes in coverage, but historical scores do not isolate either mechanism and sparse decoding is slower than dense attention. The compact 4/128-protection implementation still lacks matched quality–cost evaluation.

II. Related Work

Quest, SparQ and HISA use query-aware approximate selection to reduce the active attention set [1], [2], [3]. We adopt Quest's sign-aware page extrema for far-page scoring. AsyncTLS already combines a page-to-token cascade with coordinate selection, quantization, per-query normalization followed by GQA probability aggregation, and final attention over original KV [4]. We study these existing components with complete low-frequency RoPE pairs shared across both stages and independent near–far page quotas. Our L2 path scores the full history before masking.

Reducing attention traffic also requires considering what the selector itself reads. Token-Picker estimates probabilities from partial key bits and schedules further accesses on demand [12]; our coordinate-based scorer instead retains full-dimensional KV and derives auxiliary scoring views. KV compression and placement methods address complementary storage and movement constraints [5], [6], [7]. A smaller scoring view in our design reduces neither original-KV capacity nor, by itself, complete-request latency.

III. Near–Far Selection in a Shared Subspace

The cascade uses one coordinate map for L1 page screening and L2 token refinement (Fig. 2). L1 selects near and far pages separately. L2 masks full-history scores using this support, ranks tokens and merges the selected indices with protected positions. Final attention recomputes original-dimensional Q/K/V scores: the approximation lies in candidate selection, not in the retained logits.

A. Shared low-frequency coordinates

For Qwen3-8B, we select post-RoPE coordinates $J = \{48, \dots, 63, 112, \dots, 127\}$ from each 128-dimensional query and key. They form 16 complete low-frequency rotation pairs, not a contiguous suffix or a learned projection (Fig. 2a). With coordinate-selection matrix $E_J \in \mathbb{R}^{128\times 32}$, the scoring vectors are $z_q = E_J^\top q$ and $z_i = E_J^\top k_i$. The same map serves both regions and stages within each layer and KV head, while query heads retain their own scores. Slower positional phase variation motivates this choice; it does not guarantee retrieval fidelity. The compact implementation uses width-32 scoring tensors, whereas the historical quality runs used 32 active coordinates in width-128 L2 tensors (S7).

L1 derives page summaries and L2 derives quantize–dequantize token keys from the same coordinate map. These values remain floating point, not packed INT4. The representations use separate storage with the same coordinates in both regions. Original keys and values remain full width. The method therefore introduces auxiliary scoring representations rather than compressing the original KV cache; their preparation and maintenance belong in the total execution cost (S7).

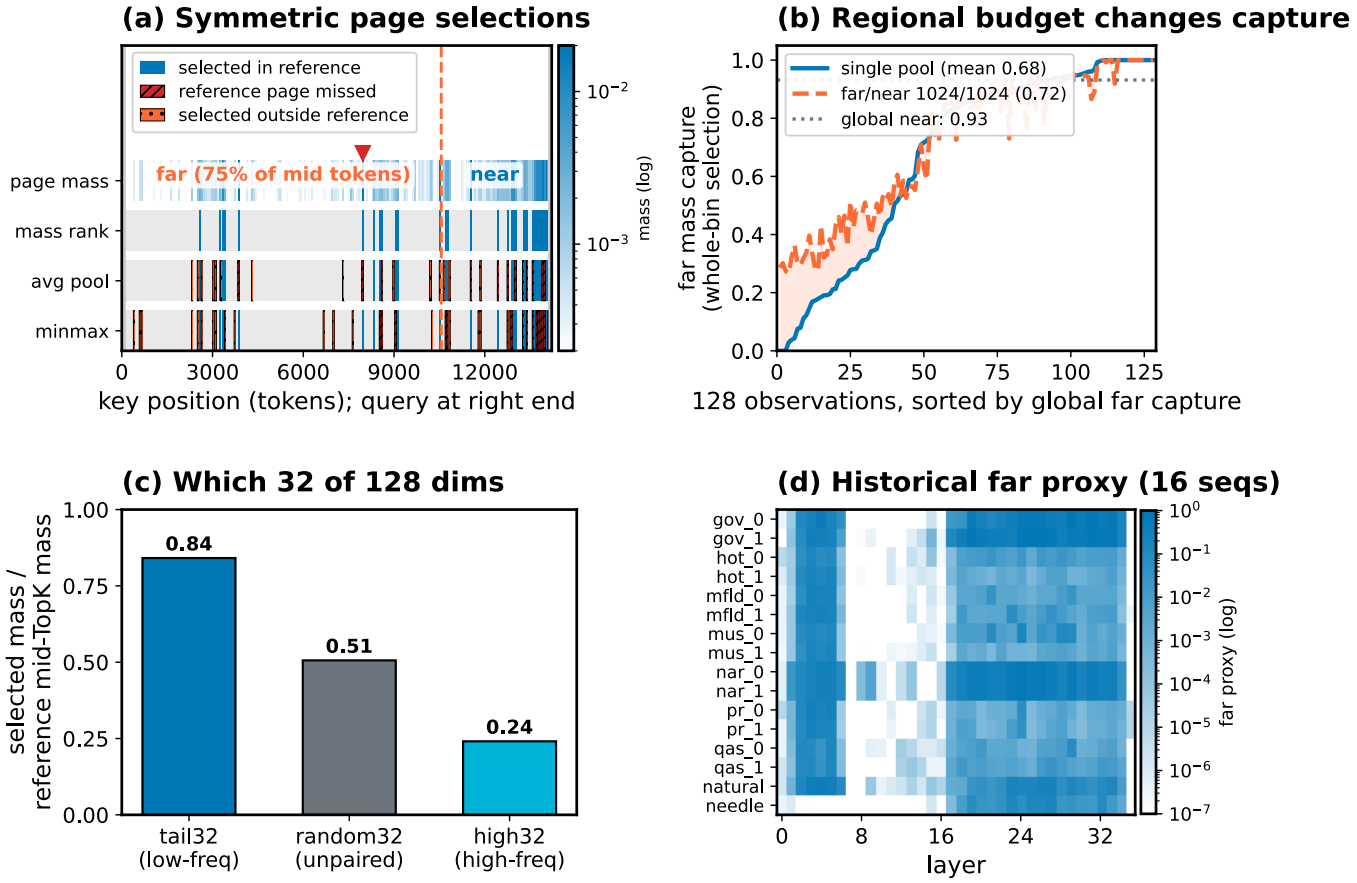


Fig. 1. Four motivating diagnostics; Qwen3-8B, 128/128 prefix/recent protection in (a)–(c). (a) MuSiQue sample 1, layer 24, query 14212. The top strip shows summed attention per page, not a selection. Below are saved mass-rank, avg and minmax sets: 32 pages each, using 1989/1989/2048 tokens under cap2048. Blue denotes selected reference pages; red hatching denotes missed reference pages; orange dots denote selected pages outside the reference. Mass-rank is a whole-page comparison reference, not an optimal oracle, exact token TopK or DSA. Set differences do not establish a quality ranking. (b) Four sequences contribute 128 correlated observations, ordered by global far retention. A global whole-bin cap of 2048 is compared with separate far/near caps of 1024/1024, without refill; red shading marks lower capture. Retention divides selected regional mass by full regional mass; the dotted line is global near retention. (c) Selected mass relative to full-dimensional reference mid-TopK mass; random32 does not preserve RoPE pairs. (d) The same 16-by-36 historical single-query proxy matrix gives 24/36 layer medians below .01 and 12–32 low-proxy layers per sequence. It uses the last sampled query, group-logit summation before softmax and far interval $[64, t-2048)$ (S6), unlike the current gate. These are diagnostic selections and profiles, not task-quality or decode-safety results.

B. Regional page and token budgets

During decoding, let t be the current token's zero-based position in the full sequence, including prefill tokens, so that $S = t+1$ counts valid cached positions, including the current position and excluding padding. Let s and w denote the protected prefix and recent-window lengths. Their union and the unprotected middle are

$$\mathcal{P}_t = \{i : 0 \le i \le t,\ i < s \text{ or } i > t - w\}, \tag{1}$$

$$\mathcal{M}_t = \{0, \dots, t\} \setminus \mathcal{P}_t. \tag{2}$$

Overlapping protection counts once. Near is a suffix of the unprotected middle history; far is its complement. Near is therefore distinct from the protected sliding window. The near boundary follows fraction α of unprotected history under minimum-page and alignment rules. The unprotected middle has length $\max(0, S - |\mathcal{P}_t|)$; the selected configuration is specified in the experimental setting.

Figure 2b details the regional cascade. Near uses average-key page scores. Far uses sign-aware extrema in the shared space. For each nonempty region-intersected page b, define $m_{b,j} = \min_{i \in b} z_{i,j}$ and $M_{b,j} = \max_{i \in b} z_{i,j}$. Quest's extrema construction [1] gives

$$U_b(z_q) = \sum_j (z^+_{q,j} M_{b,j} + z^-_{q,j} m_{b,j}), \tag{3}$$

where $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. This bounds unquantized shared-space dot products, not original-dimensional logits or separately quantized scores. Extrema can come from different tokens; a top-page shortlist can still miss important tokens. Average-key scores do not inherit this bound.

For total cap $K \ge |\mathcal{P}_t|$, define the middle-token budget $K_{\text{mid}} = K - |\mathcal{P}_t|$. Three parameters control the two-stage allocation: α sets the near-region extent, β sets its share of the L1 page cap, and γ converts retained near-page capacity into an L2 token quota. With page size B and enough pages in both regions, $P_n = \max(1, \text{round}(\beta K_1))$ and $P_f = \max(0, K_1 - P_n)$. For integer $K_1 \ge 1$, $\alpha > 0$, $0 < \beta \le 1$, $\gamma \ge 0$, and the partitioned fixed-quota path with far selection active and no quota overrides, rounding uses ties-to-even. With sufficient eligible support, let $k_n = \min(\lfloor \gamma P_n B \rfloor, K_{\text{mid}})$ and $k_f = \max(0, K_{\text{mid}} - k_n)$. Thus $P_f > 0$ does not imply $k_f > 0$: page and token access are separate constraints. For $K_1 = 128$ and $\beta = .125$, L1 reserves 16 near

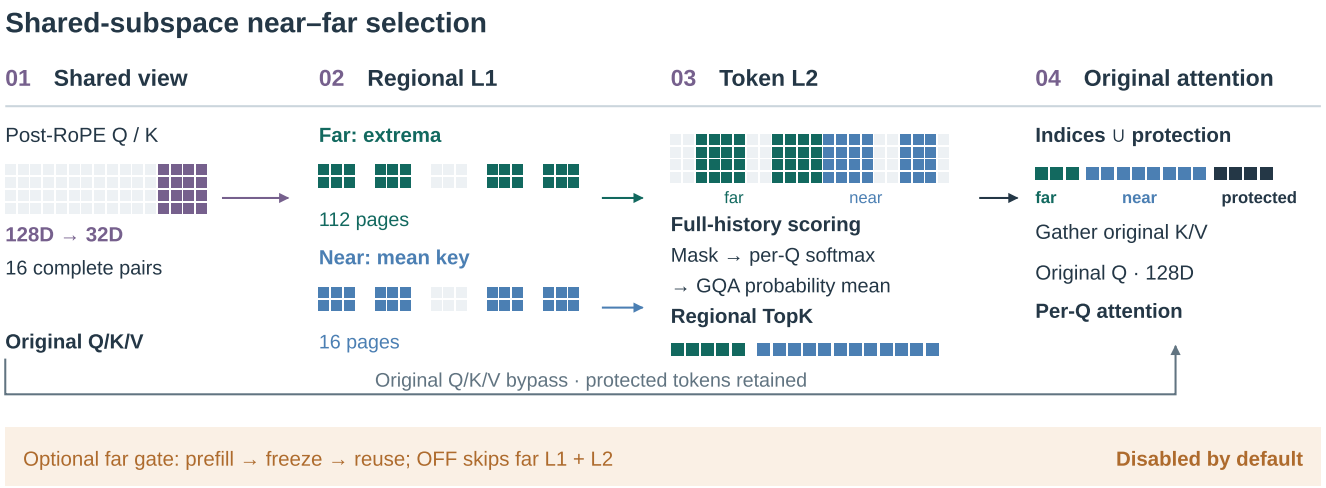


Fig. 2. Shared-subspace selection and the original-Q/K/V bypass. L1 reserves regional pages; L2 masks full-history scores, normalizes each query over shared retained support, averages probabilities within a KV group, and selects regional tokens. When OFF, the optional gate skips far scoring and redirects unused L2 quota to near candidates. Counts illustrate the sufficient-capacity configuration.

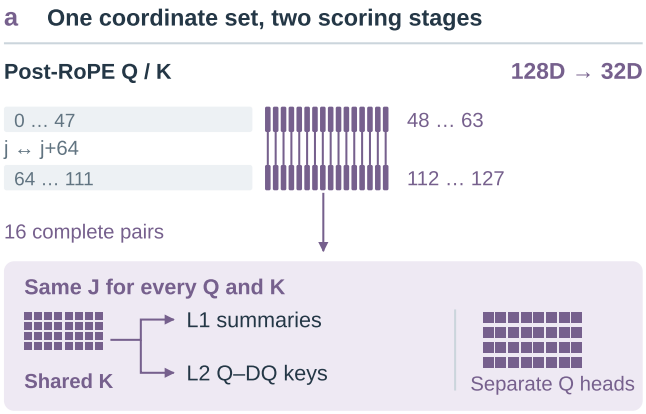


Fig. 2a. Sixteen complete split-half RoPE pairs form the 32-coordinate scoring view. Each layer and KV head derives L1 page summaries and quantize–dequantize L2 keys from the same coordinates; query views remain separate. Original Q/K/V bypass these derived views (Fig. 2).

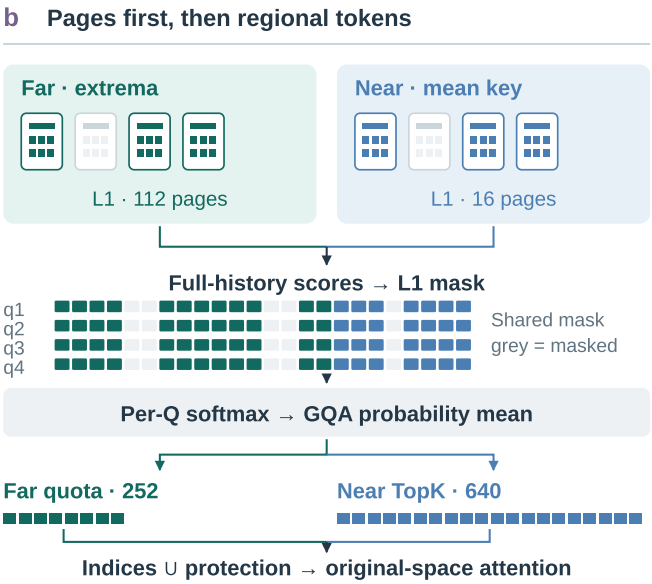


Fig. 2b. Regional screening with full-history L2 scoring. Colored cells indicate eligible candidates, not measured scores. With sufficient eligible capacity, the selected 4/128 prefix/recent configuration has 112/16 far/near pages and nominal 252/640 regional slots, plus 132 protected tokens under cap $K = 1024$. Subpage prefix reselection can consume up to four far slots. Near excludes the protected recent window.

and 112 far pages. With enough positive-probability candidates, a later token quota cannot recover a page excluded by L1. Scarce-support and masked zero-score ties are implementation boundary cases described in S7.

With sufficient disjoint history and eligible candidates, the selected configuration ($K = 1024$, $s = 4$, $w = 128$) leaves 892 middle-token slots per KV head. Here $\gamma = .625$ yields $16 \cdot 64 \cdot .625 = 640$ near and 252 far slots (Supplementary Fig. S2). Because four prefix tokens occupy only part of a page, far selection can reselect up to four protected positions; 252 is a nominal quota, not a guaranteed count of unprotected far tokens. Scarce-support and zero-tie behavior are detailed in Supplementary Note S7.

C. GQA selection and original-dimensional attention

With grouped-query attention (GQA), 32 query heads and 8 KV heads give four query heads per KV group. L1 averages their page scores before regional TopK. Ordinary L2 scores the full history for each query, applies the shared L1 mask, and averages the independently normalized query probabilities within each KV group. In this Qwen3 implementation, L2 logits retain the original-head scale $1/\sqrt{128}$ before per-query softmax, including when scoring uses 32 coordinates. This probability aggregation differs from applying softmax to averaged logits. Each query head thus retains its own approximate normalized scores before group averaging. Because scores are computed over the full history, a smaller page shortlist does not itself reduce L2 dot-product work.

The group's unique selected set $\mathcal{S}_{t,h}$ combines protection and regional winners under the total cap. All queries g in that group use these indices, but recompute their own full-dimensional attention, with head width $d = 128$:

$$o_{h,g} = \sum_{i \in \mathcal{S}_{t,h}} \frac{e^{q_{h,g}^\top k_{i,h}/\sqrt{d}}}{\sum_{j \in \mathcal{S}_{t,h}} e^{q_{h,g}^\top k_{j,h}/\sqrt{d}}} v_{i,h}. \tag{4}$$

Merging indices before softmax avoids adding separately normalized regional outputs [8]; restricting its denominator still changes attention. Physical gathering depends on the backend and length, with a dense fallback for short sequences.

D. Prefill-informed far gating

The optional gate, disabled by default, sets a layer OFF when its mean prefill far-attention weight sum is below ϵ; otherwise far selection remains active. Each input's decisions freeze at decode entry (Fig. 2). In the supported fixed-quota implementation, OFF skips both far scoring stages and their selections, while unused L2 quota refills near candidates without expanding the near-page budget. Representation preparation and sequence-wide score buffers and normalization remain.

The gate supports a single sequence (Fig. 2c; S5). Its threshold controls a prefill statistic, not subsequent error, so its quality and cost require evaluation separate from the default ungated method.

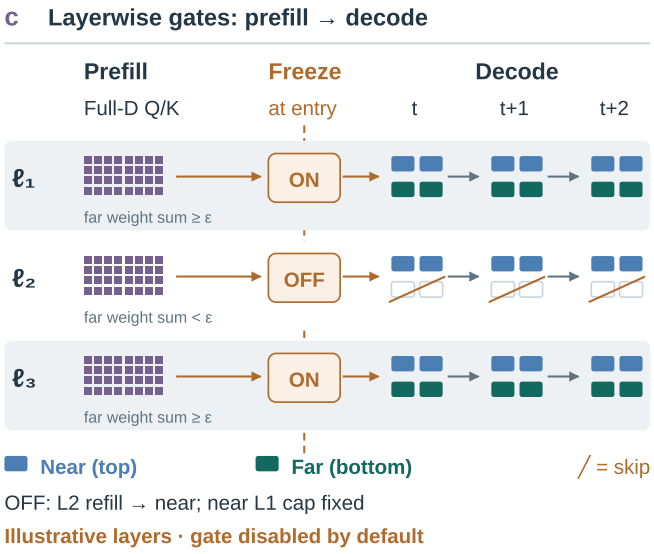


Fig. 2c. Optional far-only control, disabled by default. Full-dimensional causal prefill far-attention weight sum, averaged over the last ≤ 256 query rows and all query heads, sets one decision per input and layer. Decode entry freezes this state; new inputs reset it. OFF skips far L1/L2 and refills near L2 within unchanged near pages. Layer states are illustrative; paired cells denote L1/L2. This statistic does not bound decoding error.

IV. Experimental Setting

We evaluate three aspects separately: task quality, regional candidate coverage and execution cost. Table I defines the compact Qwen3-8B design with dense prefill, sparse decoding and a 1024-token cap including four prefix and 128 recent tokens. The available LongBench measurements instead use an earlier shared-subspace implementation: 128/128 protection and 32 active coordinates stored in zero-filled width-128 L2 keys. The comparison labels "full-dimensional page scoring" and "full-dimensional token scoring" denote reported variants whose complete execution configurations are unavailable. We retain these nominal labels without treating the runs as controlled single-stage ablations. Synthetic selector timing and model-level decode timing come from separate studies; none supplies matched quality and cost for the compact design. S6/S7 give configuration and measurement details.

TABLE I
Compact design parameters and execution policy. These settings define the proposed implementation, not the historical runs. K includes unique protected tokens.

Parameter	Setting
Final token cap K	1024
Page size; L1 page cap K_1	64; 128
α, β, γ	.25, .125, .625
L1/L2 active coordinates	32/32
Prefix / recent protection	4/128
Prefill; decode layers	dense; all sparse
L2 scoring support	full history before mask

Workloads and comparisons. The quality evidence comprises five LongBench tasks and a historical 11-task RULER [10] 128K comparison. The LongBench scores compare three nominal sparse scoring configurations; dense FullKV is a separate reference in the RULER and decode-timing records. Controlled sparse comparisons must align final unique-token cap K (including protection), model, inputs and phase policy. Realized token counts and head-sharing granularity also affect KV traffic; internal page budgets need not match (S7).

Metrics. Quality is the unweighted mean of task scores; task-specific changes are reported where available. Candidate retention measures selected attention mass divided by the corresponding full regional mass. Selector time covers a synthetic single-layer operation, whereas synchronized TPOT measures model-level intervals after the first output token. Neither is complete-request latency, which also includes preparation, maintenance and input/output work. Because the quality and timing studies are unpaired, they cannot define a quality–latency frontier. Five task means also do not establish document-paired non-inferiority; that requires matched predictions and a prespecified acceptable-loss margin.

V. Evaluation and Discussion

A. Q1: Does reduced scoring width preserve task quality?

The five-task means on HotpotQA, Qasper, MuSiQue, GovReport and RepoBench-P [9] are 47.588 for shared-subspace scoring, 47.910 for the nominal full-dimensional page variant and 47.666 for the nominal full-dimensional token variant (Supplementary Fig. S4). The shared-subspace score is lower by 0.322 and 0.078 points, respectively. These differences describe the recorded configurations; they do not show that reducing scoring width preserves quality. The variants are not verified single-factor interventions, and the shared-subspace run uses the earlier 128/128-protection implementation. Consequently, neither stage's representation effect nor the compact 4/128 design's quality can be inferred from this comparison. All three configurations are sparse; dense FullKV is a different reference.

B. Q2: How do regional budgets constrain candidate access?

Regional reservation changes which distant tokens remain reachable. In Figure 1(b), separate quotas increase mean far retention but decrease it for 43/128 observations, illustrating the cost of redistributing a fixed budget. This diagnostic uses whole-bin selection

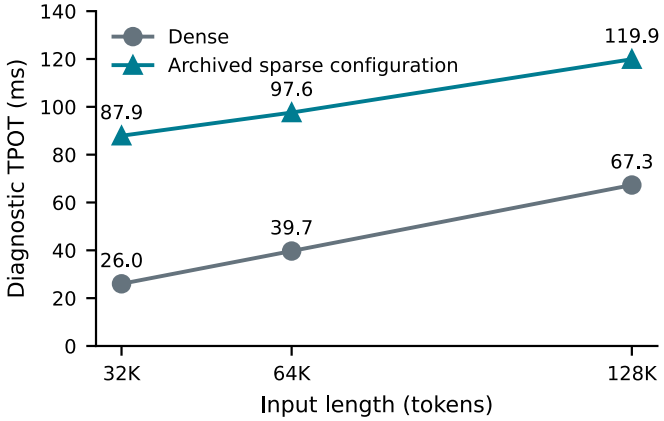


Fig. 3. Author-reported model-level decode TPOT for an archived sparse configuration and dense FullKV, Qwen3-8B with dynamic cache. Fixed 128-token output gives 127 after-first intervals, including device synchronization and argmax. The archived $(.125, .375, .625)$ setting differs from the selected 4/128 candidate. These medians do not measure complete-request E2E or establish a matched quality–cost comparison; provenance limits are in S6.

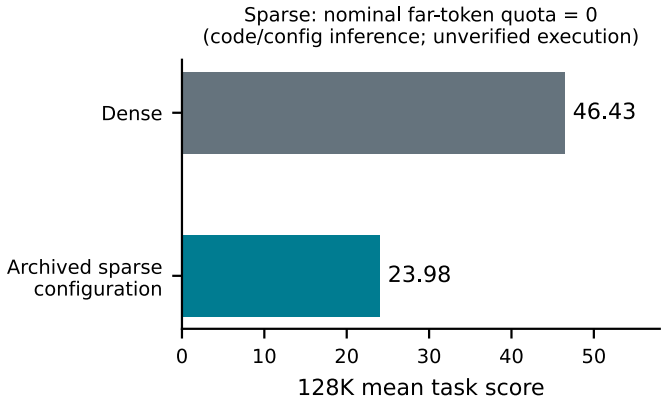


Fig. 4. Author-reported RULER 128K scores: 11 tasks, 100 examples per task and arm. The archived sparse configuration scores 22.45 points below dense FullKV; their macro-average task-score ratio is 51.6%. Reported $\beta = .375$, $K_1 = 128$ and 128/128 protection imply a nominal far-token quota of zero under the inspected formula (middle budget 768; S6). This is a code/configuration inference, not verified execution or causal attribution. The historical setting does not validate the selected 4/128 candidate or support sample-paired inference.

on four sequences, so it does not measure production-selector task quality. The two-stage budget equations identify a second constraint: even a retained far page contributes no nominal far-token slots when the near quota exhausts K_{mid}. Substituting the historical RULER setting ($\beta = .375$, $K_1 = 128$, $B = 64$, $\gamma = .625$, protection 128/128) gives $K_{\text{mid}} = 768$ and $k_f = 0$. Figure 4 reports scores of 23.98 versus 46.43 for dense attention, a 22.45-point loss, but the algebra alone neither verifies its executed path nor attributes the loss to this quota. The current compact candidate lacks a matched dense comparison.

The protection-only configuration scores 26.09 against a 47.91 reference, while using 256 rather than 1024 tokens. The difference therefore combines retrieval-policy and budget effects.

C. Q3: Does compact scoring reduce execution cost?

Model-level decoding. The reported sparse configuration is slower than dense FullKV at every measured length (Fig. 3). Sparse TPOT medians are 87.9, 97.6 and 119.9 ms at 32K, 64K and 128K, compared with dense medians of 26.0, 39.7 and 67.3 ms. The corresponding ratios are 3.38, 2.46 and 1.78. Although the relative penalty decreases with length, absolute sparse TPOT increases. Thus these measurements show no decode-time benefit from that implementation. They include device synchronization and argmax, exclude tokenization, input transfer and text output, and use a configuration different from the compact 4/128 design. Aggregate medians cannot locate the cause of the slowdown or predict the compact implementation's performance; repetition and execution-path details are in S6.

Local selector cost. A separate two-round synthetic experiment (single layer, batch one, one query, 64K QKV) reports steady-state times of 1.3502 ms for nominal full-dimensional page scoring, 1.3598 ms for shared-subspace scoring and 1.4634 ms for nominal full-dimensional token scoring. The shared-subspace selector thus has no lower reported time than the nominal page-scoring variant in this setting. An unresolved sampling-protocol discrepancy limits interpretation of small differences (S6). These timings neither locate a bottleneck nor pair with LongBench quality or complete-request cost.

D. Optional gate: prefill is not a decode guarantee

Prefill statistics also fail to bound later far-region demand: an observed input–layer pair with prefill far-attention weight sum below .05 reaches .1678 during decoding. Relative to the reported reference score of 47.910, gate thresholds of .05 and .10 are associated with mean decreases of .034 and .218 points, respectively; three of five tasks decline in each configuration, including HotpotQA decreases of 1.05 and 1.26 points. The larger per-task losses demonstrate that a small average change can conceal task sensitivity.

The gate summaries omit the complete scoring configuration, precluding attribution to gated shared-subspace selection (S6).

Design implications and limitations. Three constraints emerge. First, scoring width controls the representation used for ranking, but the current L2 path still performs full-history scoring and normalization. Second, regional page access and regional token access require separate budgets; satisfying the first does not ensure the second. Third, the cost of preparing and using a selector must be recovered by reduced attention work. The reported timings do not show that recovery. The missing matched compact evaluation prevents a quality–cost conclusion, and the historical gate summaries do not identify the shared-subspace configuration. The evidence therefore supports the mechanism and these constraints, while leaving the proposed combination's comparative benefit unresolved.

VI. Conclusion

We studied sparse attention as a joint representation and allocation problem. A shared low-frequency coordinate map supplies page and token scoring, regional quotas control access to refinement, and final attention uses original-dimensional Q/K/V. The analysis distinguishes low-dimensional ranking from reduced execution work, and far-page reservation from final far-token coverage. Qwen3 diagnostics illustrate these trade-offs, including regional-retention losses, substantial historical quality loss and slower sparse decoding. Establishing a useful operating point for the compact 4/128 design requires measuring quality and preparation-inclusive cost under the same configuration; the available separate studies do not yet establish that benefit.

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